Investment in Demand and Dynamic Competition for Customers

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- **Data:** Firms compete for customers with **prices** and with **investment in demand**
  - Number of customers accounts for >75% of sales variation across firms
    (Bernard-Dhyne-Magerman-Manova-Moxnes '22, Einav-Klenow-Levin-MurcianoGoroff '22, Afrouzi-Drenik-Kim '25)
  - **Sales & marketing** to contact & inform customers $\sim 2/3$ the size of K investment (He-Mostrom-Sufi '25)
  - Customers **select suppliers by price** $\rightarrow$ competition for customers (Paciello-Pozzi-Trachter '19, Gilbukh-Roldan '21)

- **Theory:** Prominent macro benchmarks isolate one margin of price competition
  - Static demand: **intensive** margin only (Dixit-Stiglitz '77, Baqaee-Farhi '20, Edmond-Midrigan-Xu '23)
  - Search frictions: **extensive** margin only (Gourio-Rudanko '14, Menzio '26)
  - Customers "produced" by demand investment: **intensive** margin only (Afrouzi-Drenik-Kim '25)

- **This paper:** Prices act at **both margins**; demand investment sets **how many offers** customers see
  $\Rightarrow$ Changes what markups **cost**, whether firms **over-invest**, and the right **policy mix**

This Paper: Setup and Main Takeaways

- **Quantitative theory** of investment in demand and dynamic competition for customers
  - Product-market search frictions $\rightarrow$ investment in demand = cost of contacting customers
  - Customers choose suppliers & quantity/supplier by price $\rightarrow$ **extensive** & intensive margin competition
  - $\Rightarrow$ Consistent with evidence on sales, markups, and investment in demand

- **Main takeaways:** Both margins matter, and they **interact** $\rightarrow$
  1. *Pricing:* **source** $\neq$ **cost** of markups: extensive margin disciplines them, intensive margin sets the cost
     - $\Rightarrow$ A given markup dispersion is $> 3\times$ **less costly** than if it came from product differentiation alone
  2. *Investment:* firms ignore **customer surplus** (too little) and the **business they steal** (too much)
     - $\Rightarrow$ Business stealing dominates: equilibrium **over-invests**, planner cuts aggregate demand investment by $\sim 1/3$
  3. *Interaction:* investment tax softens price competition, production subsidy fuels investment
     - $\Rightarrow$ The two instruments are **complements**

- **Application:** Rising demand investment $\rightarrow$ sales concentration $\uparrow$ **without** more market power

Related Literature

- Markups and **misallocation of demand**
  - **Static Demand:** Restuccia-Rogerson ('08), Hsieh-Klenow ('09), Dhingra-Morrow ('19), Baqaee-Farhi ('20), Peters ('20), **Edmond-Midrigan-Xu ('23)**, deRidder ('24), Baqaee-Farhi-Sangani ('24a,b), Bornstein-Peter ('25), Li-Lian-Ma-Martell ('25), Becker-Edmond-Midrigan-Xu ('26) ...

Here: Markups from **competition for customers**, not product differentiation → source ≠ cost of markups

  - **Dynamic Demand:** Cavenaile-Celik-Perla-Roldan ('26), **Afrouzi-Drenik-Kim ('25)**, Cavenaile-Celik-Roldan-Tian ('25), Pearce-Wu ('25)

Here: **Prices** and ongoing demand investment jointly determine the evolution of **customer relationships**

- **Search frictions** in product markets
  - Stigler ('61), **Butters ('77)**, Burdett-Judd ('83), Grossman-Shapiro ('84), Dinlersoz-Yorukoglu ('12), Drozd-Nosal ('12), Gourio-Rudanko ('14), Paciello-Pozzi-Trachter ('19), Gilbukh-Roldan ('21), Albrecht-Menzio-Vroman ('23), Menzio ('23), Boehm-Oberfield-South-Waseem ('24), Chiavari ('24), Greenwood-Ma-Yorukoglu ('24), Menzio ('24), Rudanko ('25), Shen ('25), **Menzio ('26)**, ...
  - **Efficiency in Search Markets:** Mortensen ('82), Hosios ('90), Fukui-Mukoyama ('25),...

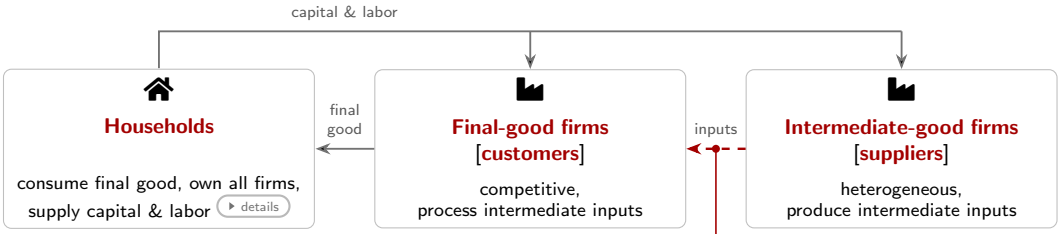
Here: Quantify misallocation from markups, **over-investment** in demand (poaching), and their **interaction**

- **Evidence** on sales concentration, competition, customer dynamics, and sales & marketing
  - Autor-Dorn-Katz-Patterson-VanReenen ('20), Barkai ('20), DeLoecker-Eeckhout-Unger ('20), Bernard-Dhyne-Magerman-Manova-Moxnes ('22), Einav-Klenow-Levin-MurcianoGoroff ('22), Fitzgerald-Haller-YedidLevi ('24), Argente-Fitzgerald-Moreira-Pirolo ('25), **He-Mostrom-Sufi ('25)**, ...

Here: New quantitative theory to jointly account for demand investment, sales, pricing & markup patterns

Framework

Environment



Frictional market for intermediates

- Customers can only purchase if in contact with a supplier
- Persistent matches between customers and suppliers
- Suppliers **cannot commit** to future prices
- Customers can switch supplier → **poaching**
- Customers observe only prices → **Bayesian equilibrium**

Final-Good Firms (Customers)

- **Competitive** final-good firms combine intermediates j to produce final good Y

$$\textbf{CES technology:} \quad Y = \left(\int_0^1 y_j^{\frac{\sigma-1}{\sigma}} dj \right)^{\frac{\sigma}{\sigma-1}} \quad \Rightarrow \quad \textbf{Demand for } j\textbf{:} \quad y_j = (mc_j)^{-\sigma} Y$$

- Cost mc_j of using intermediate input j:
  - If no supplier for j (**unmatched**): in-house production $y_j = k_j^{\alpha} \ell_j^{1-\alpha} \rightarrow mc_j = \overline{mc}$
  - If **matched** to a supplier charging p: procure y_j externally $\rightarrow mc_j = p$

- Static **value of a match** = reduction in production cost $\rightarrow \eta(p) = \frac{p^{1-\sigma} - \overline{mc}^{1-\sigma}}{\sigma - 1} Y$

- Persistent matches $\rightarrow$ dynamic values of being matched $M(p)$ or unmatched U
  - $M(p)$ accounts for price changes & exit of current supplier, rematching with new supplier
  - **No commitment**: can **switch supplier if contacted** or always quit to produce in-house

▸ dynamic ▸ industries

Intermediate-Good Firms (Suppliers)

- For each j: Continuum of suppliers **differ in initial customers** n and stochastic **productivity** z
- **No commitment** & no price discrimination $\rightarrow$ set one price p for all customers each period
  - Produce with technology $y = zk^{\alpha}\ell^{1-\alpha}$ $\rightarrow$ marginal cost $\frac{\overline{mc}}{z}$
  - **static profit** per customer $\pi(p, z) = \left(p - \frac{\overline{mc}}{z}\right) y(p)$
- Choose how many **new customers** $\hat{n}$ to contact, subject to cost $\hat{s}(\hat{n}, n)$ $\rightarrow$ **demand investment**

Customer growth: $n' = n + e(p)\hat{n} - q(p)n$

  - **Conversion rate** $e(p)$: new customers per contact
  - **Quit rate** $q(p)$: separations per existing customer

} **equilibrium objects**

$\Rightarrow$ Variation in both **extensive** (n') and **intensive** ($y(p)$) margin of demand

  - Consistent with empirical evidence on sales variation

(Bernard-Dhyne-Magerman-Manova-Moxnes, '22; Einav-Klenow-Levin-MurcianoGoroff, '22; Afrouzi-Drenik-Kim, '25; Argente-Fitzgerald-Moreira-Pirolo, '25)

Intermediate-Good Firms: Dynamic Problem

- Assume $\hat{s}(\cdot,\cdot)$ is **constant returns to scale** $\rightarrow \hat{s}(i \cdot n, n) = s(i)n = \frac{\bar{s}}{\psi} i^{\psi} n$
  - Define $i = \frac{\hat{n}}{n}$ as **contact intensity** $\rightarrow$ growth rate $\frac{n'}{n} \equiv \rho(p,i) = 1 - q(p) + e(p)i$

- Production firms' value $\tilde{V}(z,n)$ **linear in** $n \rightarrow \tilde{V}(z,n) = nV(z)$

$$V(z) = \max_{p,i} \quad -s(i) + \rho(p,i) \left(\pi(p,z) + \frac{1-\delta_F}{1+r} \mathbb{E}\left[\max\left\{ V\left(z'\right), 0 \right\} | z \right] \right)$$

$\Rightarrow$ **Optimal choices** $p(z)$ and $i(z)$ depend only on z, **independent of** n
  - Consistent with firms not varying prices with # customers across markets
    (Fitzgerald-Haller-YedidLevi, '24; Argente-Fitzgerald-Moreira-Pirolo, '25)
  - Despite responsiveness of customer growth to price $\rightarrow$ **choice, not constraint**
    (Paciello-Pozzi-Trachter, '19; Gilbukh-Roldan, '21)

Intermediate-Good Firms: Optimal Choices

- The optimal price $p(z)$ trades off **profits per customer** against **number of customers**

$$\underbrace{\rho(p,i)\frac{\partial \pi(p,z)}{\partial p}}_{\text{effect on current profit}} = -\underbrace{\left[-\frac{\partial q(p)}{\partial p}+\frac{\partial e(p)}{\partial p}i\right]}_{\substack{\text{effect on}\\ \text{customer growth}}}\underbrace{\left(\pi(p,z)+\frac{1-\delta_F}{1+r}\mathbb{E}\left[V\left(z'\right)|z\right]\right)}_{\text{value per additional customer}}$$

- Optimal contact intensity $i(z)$ trades off **cost of contacts** against **customer growth**

$$\frac{\partial s(i)}{\partial i} = \underbrace{e(p)}_{\substack{\text{effect on}\\ \text{customer growth}}}\underbrace{\left(\pi(p,z)+\frac{1-\delta_F}{1+r}\mathbb{E}\left[V\left(z'\right)|z\right]\right)}_{\text{value per additional customer}}.$$

- **Competitors'** choices affect optimal price and contact intensity through $q(p)$ & $e(p)$

Matching of Customers and Suppliers

- Suppliers contact customers at **random** $\rightarrow$ contacts per customer **Poisson**-distributed with mean

$$I = \int i(z) dG^i(z)$$

- Customers observe prices of existing supplier and new contacts $\rightarrow$ choose highest value $M(p)$

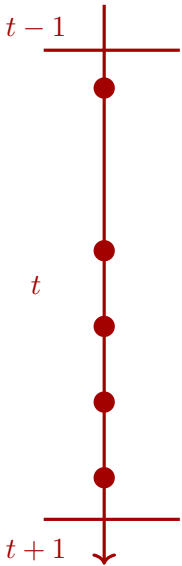
$\Rightarrow$ Distribution of contacts & customers' choices determine **quit rate** $q(p)$ and **conversion rate** $e(p)$

$\Rightarrow$ When posting prices, suppliers form **expectations about potential competitors**

- How many other contacts do customers have? What prices do my competitors charge?
- Price set based on expectation of being undercut $\rightarrow$ **"limit-pricing in expectation"**

▸ timing

Timing, Entry & Exit



1. **Productivity** realizes, suppliers **enter & exit**
- Exog. (δ_F) & endog. ($V(z)<0$) exit $\rightarrow$ customers move to unmatched
- Entrants contact unmatched w/ prob. $\chi \rightarrow$ spin-off in-house production
- Draw initial z from $F_0(z)$; enter iff $V(z) \geq 0$

2. Active **suppliers choose** contact intensity i and price p

3. Suppliers **contact** $i \cdot n$ customers at random

4. **Customers decide**: stay/relocate/quit to unmatched & how much to use input j

5. **Production** and trade

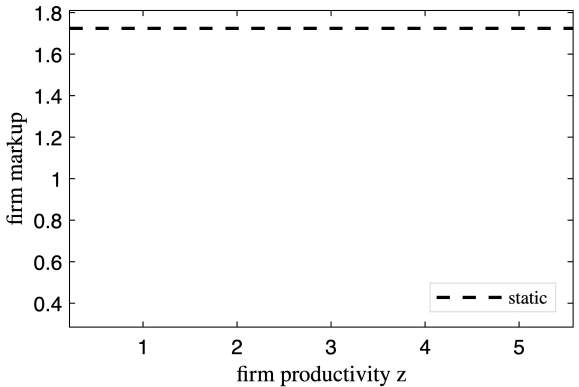
Constructing an Equilibrium

- Solve for **Bayesian equilibrium** in pure Markov strategies (Coles-Mortensen '16)
  - Fixed point in firms' policies $p(z)$ and $i(z)$, customer beliefs $H(p'|p)$, customer distribution $G^i(z)$
- Focus on **separating equilibria** with prices strictly decreasing in productivity z
  - Additional restriction on monotonicity of off-equilibrium beliefs
- Considered equilibria share **four properties**:
  1. Customers quit to unmatched above **reservation price** $\bar{p}$
  2. Conditional on contacts, customers **choose highest** z **supplier** (Menzio, '26)
  3. No firm charges above $p^S(z) = \arg\max \pi(p, z)$
  4. Any equilibrium pricing function $p(z)$ is continuous and has $p(\underline{z}) = \bar{p}$

▸ definition ▸ eqm. $\rho(p, i)$ ▸ firm FOCs

Three Forces Determine Markups

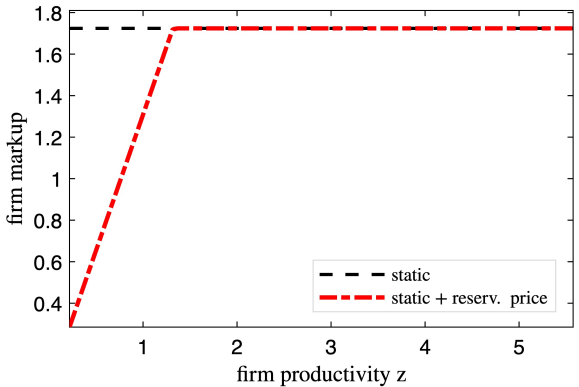
1. **Intensive-margin**: elasticity σ determines **static monopoly markup**
2. **Extensive-margin**: outside option $\Rightarrow$ reservation price $\bar{p}$ lowers markup in left tail
3. **Extensive-margin**: others' demand investment $\uparrow \Rightarrow$ ext.-margin elasticity $\uparrow \Rightarrow$ eqm. markup $\downarrow$



own i $p(z)$

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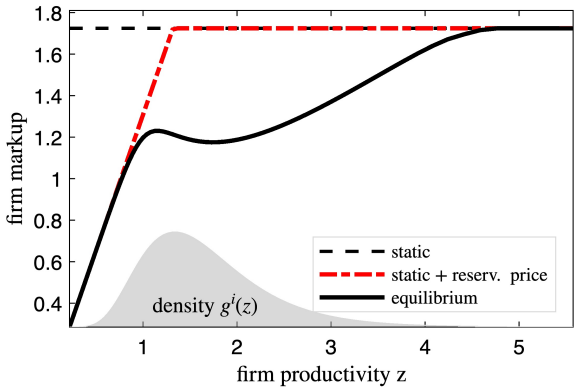
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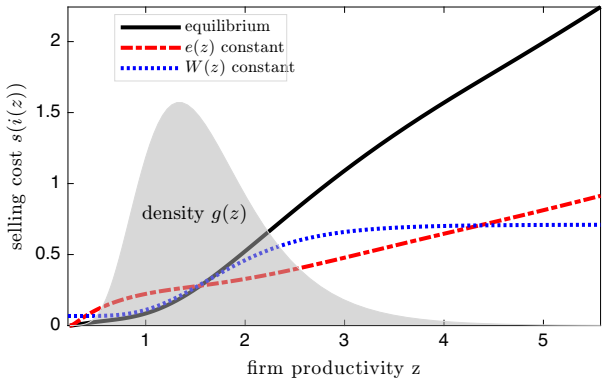
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Equilibrium Demand Investment

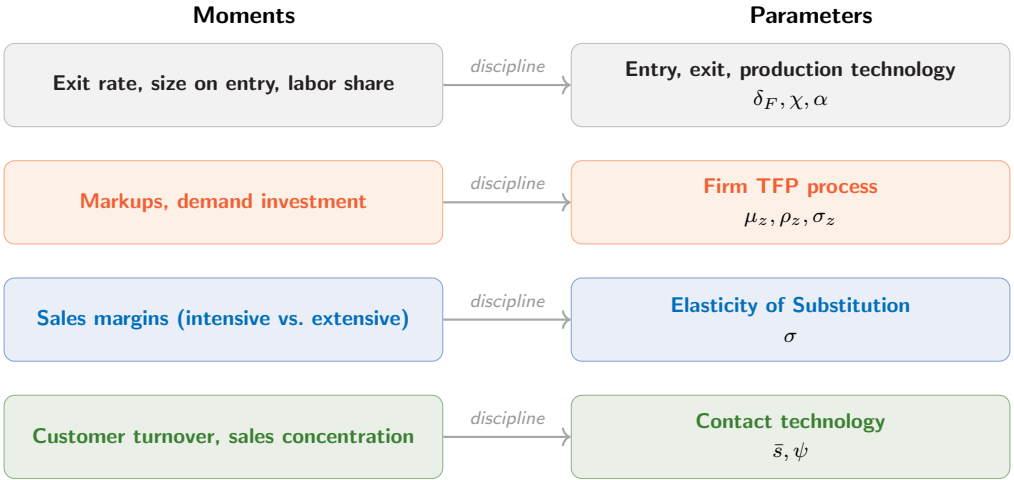
- Contact intensity **increases in productivity**, due to two forces:
  1. Increasing value (**profits**) per customer: $W(z) = \pi(p(z), z) + \frac{1-\delta_F}{1+r}\mathbb{E}\left[\max\{V(z'), 0\}|z\right]$
  2. Increasing gains in # of customers: **conversion rate** $e(p(z))$



▸ growth rate decomposition

Quantification

Model Disciplined by Cross-Section of Sales, Markups & Demand Investment



Calibration & Model Fit

Targeted moment (10)	Model	Data	Source	Parameter (9)		Value
Firm entry/exit rate	0.081	0.085	BDS	δ_F	Exogenous firm exit rate	0.081
Relative employment of entrants	0.526	0.540	Sterk-Sedlacek-Pugsley ('21)	χ	Entrants per unmatched customer	0.078
Aggregate labor share	0.614	0.615	BLS	α	Capital elasticity	0.327
Aggregate markup	1.250	1.250	Capital IQ	σ_z	Log TFP: unconditional dispersion	0.387
Elasticity of $\frac{\text{demand investment}}{\text{revenue}}$ to markup	1.849	1.843	Capital IQ, He-Mostrom-Sufi ('25)	ρ_z	Log TFP: auto-correlation	0.361
Elasticity of markup to revenue	0.031	0.031	Edmond-Midrigan-Xu ('23)	μ_z	Log TFP: mean	0.409
Sales variation: intensive-margin share	0.262	0.260	Bernard-Dhyne-Magerman-Manova-Moxnes ('22)	σ	Intensive-margin elasticity	2.381
Customer quit rate (% sales)	0.152	0.150	Dhyne-Duprez-Komatsu ('23)	$\bar{s}$	Contact cost: level	39.792
Top 10% sales share	0.676	0.724	SUSB	ψ	Contact cost: curvature	2.895
Top 1% sales share	0.431	0.423	SUSB			

Parameters are calibrated jointly; each parameter is listed next to its most informative moment.

▸ investment vs. markup ▸ markup distribution ▸ moments by firm revenue ▸ i by industry ▸ robustness ▸ sensitivity

Model Validation

- Calibration consistent with additional evidence on competition and industry dynamics
  - σ consistent with direct estimates of **elasticity of substitution** for intermediate inputs
  - **Markups** are correlated with sales/customer, but not # of customers
  - Generate realistic **customer surplus** (cost reduction) from additional supplier $\eta(p)$

	Model	Data	Source
Elasticity of substitution σ across intermediate inputs	2.38	0.7–2.5	Carvalho-Nirei-Saito-TahbazSalehi ('21) Peter-Ruane ('25)
Elasticity of markup to # of customers	0.000	−0.004	Afrouzi-Drenik-Kim ('25)
Elasticity of markup to revenue per customer	0.152	0.187	Afrouzi-Drenik-Kim ('25)
Customer surplus (rel. to expenditure)	0.30	0.28	Baqaee-Burstein-Duprez-Farhi ('25)
Log revenue auto-correlation	0.990	0.991	Capital IQ
Demand investment as share of GDP	0.113	0.080	Capital IQ
Cost pass-through	0.799	0.59–0.76	Amiti-Itskhoki-Konings ('19)

Efficient Allocation & Product Market Policies

Social Planner: Setup

- Planner faces **same search frictions & technologies**, makes time-varying choices for all agents:
  1. Aggregate consumption and supply of labor & physical capital
  2. Quantity of intermediate inputs conditional on matches
  3. Investment in demand & supplier exit $\rightarrow$ distribution of customers across suppliers

- Equilibrium is **inefficient** for three reasons:
  1. **Cost of aggregate markup:** positive aggregate markup distorts factor supply
  2. **Intensive margin misallocation:** dispersion in markups distorts intermediate input choice
  3. **Extensive margin misallocation:** planner values contacts (matches) differently than suppliers

▸ problem ▸ agg. μ ▸ disp. μ

Social Planner: Extensive Margin Misallocation

- Direct externality of contacts on **matches of other firms**
  - **Poaching** from existing matches & **crowding out** of matches with alternative contacts

$$\frac{\partial s(i(z))}{\partial i} = e^*(z)W^*(z) + \underbrace{\int_0^\infty W^*(\hat{z})\frac{g^{i*}(\hat{z})}{g^{i*}(z)}\left[-\frac{\partial q^*(\hat{z})}{\partial i^*(z)} + \frac{\partial e^*(\hat{z})}{\partial i^*(z)}i^*(\hat{z})\right]d\hat{z}}_{\text{static business stealing}}$$

- **Value of a match** not aligned for planner and supplier
  - Existing matches affect future **quit** and **conversion** rates of other suppliers & **future entry**
  - Total private benefit includes **customer surplus** ($\neq \pi(p(z), z)$)

$$V^*(z) = -s(i^*(z)) + (1 - q^*(z) + e^*(z)i^*(z))W^*(z) + \underbrace{\int_0^\infty W^*(\hat{z})g^{i*}(\hat{z})\left[-\frac{\partial q^*(\hat{z})}{\partial g^{i*}(z)} + \frac{\partial e^*(\hat{z})}{\partial g^{i*}(z)}i^*(\hat{z})\right]d\hat{z}}_{\text{dynamic business stealing}}$$

$$W^*(z) = \underbrace{\frac{\partial Y^*}{\partial g^*(z)}}_{\substack{\text{private benefit} \\ \text{(supplier \& customer)}}} + \beta\int_0^\infty (1 - \delta^*_{t+1}(\hat{z}))V^*_{t+1}(\hat{z})f(\hat{z}|z)d\hat{z} - \underbrace{\beta\chi\int_0^\infty (1 - \delta^{*E}_{t+1}(\hat{z}))V^*_{t+1}(\hat{z})f_0(\hat{z})d\hat{z}}_{\text{dynamic business stealing}}$$

Decentralized Equilibrium vs. Social Planner

- Planner maximizes lifetime utility, generates **aggregate welfare gain** of 5.1% along transition
  - Eliminating the aggregate markup increases L and K → welfare $+1.9\%$
  - Eliminating markup dispersion resolves intensive margin misallocation → welfare $+2.2\%$
  - Extensive margin misallocation: planner cuts demand investment by 36.9% → welfare $+1.0\%$

	Eqm.	Planner		Aggr. Markup	Markup Disp.	Dist. of Matches
Steady state						
Output	1.09	1.29	$+17.7\%$	$+20.8\%$	$+2.7\%$	-5.7%
Physical capital	2.93	4.13	$+40.9\%$	$+44.5\%$	$+3.2\%$	-6.8%
Labor	0.33	0.35	$+8.9\%$	$+10.7\%$	-0.1%	-1.7%
Consumption	0.82	0.98	$+18.9\%$	$+18.0\%$	$+2.9\%$	-2.0%
Aggregate TFP	1.64	1.63	-0.6%	0.0%	$+1.6\%$	-2.2%
Demand investment	0.09	0.06	-36.9%	0.0%	0.0%	-36.9%
Welfare (CEV, steady state)			$+11.0\%$	$+8.5\%$	$+2.7\%$	-0.3%
Welfare (CEV, incl. transition)			$+5.1\%$	$+1.9\%$	$+2.2\%$	$+1.0\%$

Note: Equilibrium GDP (= output − demand investment) is normalized to 1.00.

▸ full ▸ fix $G(z)$ ▸ fix $G(z)$ & Z

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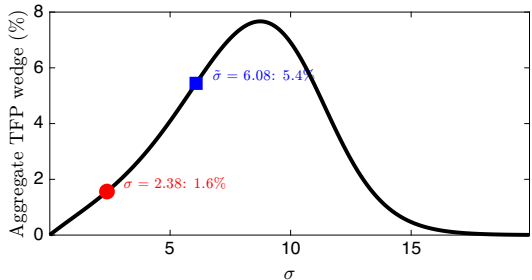
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Consumption	0.82	0.98	$+18.9\%$	$+18.0\%$	$+2.9\%$	-2.0%
Aggregate TFP	1.64	1.63	-0.6%	0.0%	$+1.6\%$	-2.2%
Demand investment	0.09	0.06	-36.9%	0.0%	0.0%	-36.9%
Welfare (CEV, steady state)			$+11.0\%$	$+8.5\%$	$+2.7\%$	-0.3%
Welfare (CEV, incl. transition)			$+5.1\%$	$+1.9\%$	$+2.2\%$	$+1.0\%$

Note: Equilibrium GDP (= output − demand investment) is normalized to 1.00.

▸ full ▸ fix $G(z)$ ▸ fix $G(z)$ & Z

Markups and Misallocation: Role of σ

- We find **comparably small misallocation** from markup dispersion (aggr. TFP $1.6\% \downarrow$)
- **Determinants:** magnitude of dispersion vs. cost of given dispersion (intensive-margin elasticity σ)
  - Standard model (e.g. Kimball): σ used to match $\bar{\mu} \rightarrow$ markups & cost not independent
  - Here: markups determined by search friction $\rightarrow$ σ **free** to match variation in sales/customer
  - We calibrate **low** $\sigma \approx 2.4$, targeting **low intensive-margin sales variation**
- **Counterfactual:** Hold markups and customer allocation fixed; vary only σ
  - Matching $\bar{\mu} = 1.2$ through intensive demand alone: $\tilde{\sigma} = 6.08 \rightarrow$ TFP loss $1.6\% \rightarrow 5.4\%$ $(> 3\times)$

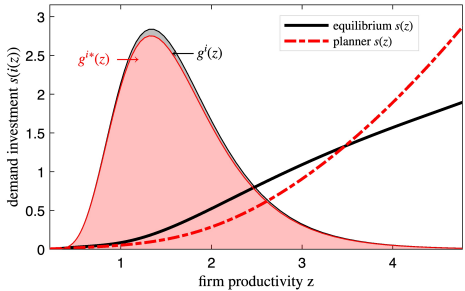


▸ alt. model

Equilibrium vs. Planner: Demand Investment

- Planner reduces **aggregate** demand investment by **37%** on net, reflecting competing forces
  - **Business stealing** externalities dominate private benefit to customer
  - Quantitative magnitudes of forces consistent with empirical estimates
    (Baqaee-Burstein-Duprez-Farhi, '25; Sinkinson-Starc, '19)

- Planner reallocates demand investment: **increases** it on net for top **2%** of firms (by z)
  - Relative strength of externalities differs across z, private benefit dominates in right tail



▸ decomp. ▸ $\bar{s}$ ▸ validation

Optimal Product Market Policy

- Maximize welfare (transition) with flat **tax on investment** τ_S and flat **production subsidy** τ_P
  - Common across all $z \to$ target average inefficiency, not implementation of full planner solution
  - Budget balanced period-by-period with lump-sum transfer (tax) to household
- Investment tax and production subsidy are **complements**
  - When considered jointly: higher tax & subsidy, more than additive welfare gain

% change vs. no-tax steady state	Production subsidy	Investment tax	Both
Investment tax τ_S	–	33.8%	61.1%
Production subsidy τ_P	13.3%	–	17.1%
Demand investment	+22.4%	−32.8%	−31.6%
Aggregate Markup	−0.3%	+0.6%	+0.5%
Aggregate TFP	+1.2%	−2.1%	−2.0%
Welfare (CEV) – transition	+1.3%	+0.5%	**+2.5%** – 1.3% + 0.5%

▸ full ▸ size ▸ τ_P ▸ τ_S ▸ both ▸ robustness

Macroeconomic Consequences of Rising Demand Investment

Rising Demand Investment, Concentration, and Markups

- Since 1980s, joint increase in sales concentration and demand investment
  - Depending on measure, demand investment has increased by 15–35% as a share of revenue
    (Kaplan-Zoch, '24; Bronnenberg-Dube-Syverson, '22; Greenwood-Ma-Yorukoglu, '24; KLEMS)
  - At same time, sales concentration has increased by $\sim 25\%$
    (Autor-Dorn-Katz-Patterson-VanReenen '20)

- In model, both trends rationalized by an **improvement in contact technology**
  - Target 15% increase in demand investment $\rightarrow$ infer that average contact costs fell by about 31%
  - **Better customer allocation** raises TFP by 3.1% and GDP by 4.6%

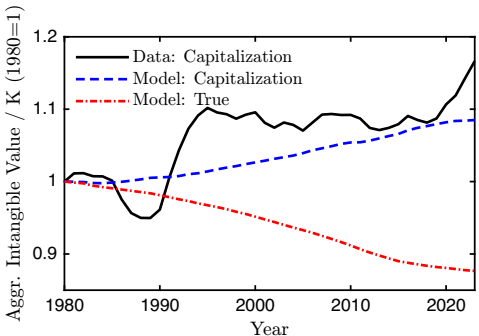
- Higher sales concentration without higher aggregate markup $\rightarrow$ **concentration $\neq$ competition**
  - Offsetting forces: reallocation to high markup firms, but lower markup for given firm
    (DeLoecker-Eeckhout-Unger, '20)
  - **Agg. markup increase** can be rationalized via higher productivity dispersion σ_z

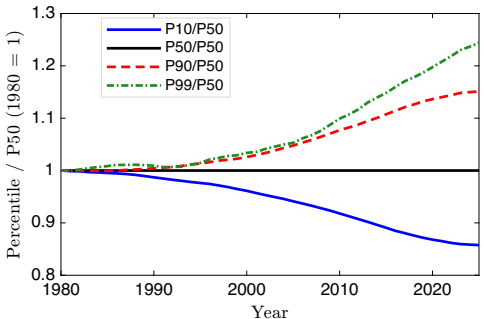
▸ data ▸ table ▸ markup exercise

Demand Investment and Intangible Capital

- Decline of intangible (customer) capital despite higher demand investment
  - **Capitalized** intangible value (const. depreciation rate) ↑ vs. **true intangible intensity** ↓
  - Difference reflects **equilibrium effect** of increased poaching → **endogenous "depreciation"** ↑

- **Fanning-out** of (intangible) firm values, consistent with evidence (Eeckhout, '25)
  - Increased poaching hurts less productive firms, but allows more productive firms to grow faster





▸ intangibles ▸ firm values

Conclusion

Conclusion

- **Quantitative theory** of investment in demand and dynamic competition for customers
  - Fits salient facts on sales, markups, and investment in demand
  - Key feature: **interaction** of competition in customer acquisition and pricing

- **Main takeaways:**
  1. *Pricing:* source $\neq$ cost of markups $\rightarrow$ **lower cost of given markup dispersion**
     - TFP loss of 1.6%, vs. 5.4% if the same markups came from product differentiation
  2. *Investment:* Business stealing dominates customer surplus $\rightarrow$ **over-investment**
     - Planner cuts demand investment by 37%
  3. *Interaction:* Investment tax softens price competition; production subsidy fuels investment
     - **Complementary instruments**: welfare $+2.5\%$ jointly, vs. $+0.5\%$ (tax) and $+1.3\%$ (subsidy) alone

- **Application:** Rising demand investment $\rightarrow$ sales concentration $\uparrow$ **without** more market power
  - Model rationalizes both trends with an improvement in contact technology

Appendix

Households

- The **representative household** has preferences over final consumption C and labor L
- Saves in physical capital K at return r, provides labor at wage w
- Owns all firms and receives profits Π

$$
\begin{aligned}
V(K) = \max_{C,K',L} \; & u(C) - v(L) + \beta V(K') \\
\text{s.t.} \quad & C + K' = wL + (1+r)K + \Pi \\
& K' \geq 0
\end{aligned}
$$

back

Dynamic Value of Supply Relationship: Production Stage

- Value of a match with a firm charging p at production stage
  - $H(z|p)$ is customers' belief about productivity and $\gamma(p) = \delta_F + (1-\delta_F)\int_0^\infty F(\underline{z}|z)dH(z|p)$

$$M(p) = \max\left\{\eta(p) + \frac{1}{1+r}\left[(1-\delta_F)\int_0^\infty \int_{\underline{z}}^\infty M^i(p(z'))dF(z'|z)dH(z|p) + \gamma(p)U^i\right], U\right\}$$

- Value of being unmatched at production stage
  - χ is the likelihood an unmatched division is contacted by an entrant

$$U = \frac{1}{1+r}\left[U^i + \chi\int_{\underline{z}}^\infty \left(M^i(p(z)) - U^i\right)dF_0(z)\right]$$

back

Dynamic Value of Supply Relationship: Re-Matching Stage

- Value of a match with a firm charging p at re-matching stage

$$M^i(p) = M(p) + \underbrace{\int_0^\infty \max\{M(p(z)) - M(p), 0\}\, dO(z)}_{\text{expected gain from finding supplier}}$$

- Value of being unmatched at re-matching stage

$$U^i = U + \underbrace{\int_0^\infty \max\{M(p(z)) - U, 0\}\, dO(z)}_{\text{expected gain from finding supplier}}$$

- Here $O(z) = e^{-I(1-D(z))}$ is the Poisson order-statistic CDF: probability that the highest-productivity contact has productivity $\leq z$

$\Rightarrow$ $M^i(p(z)) \geq U^i$ $\forall z \geq \underline{z}$ $\rightarrow$ no customer leaves a continuing firm before re-matching

back

Stationary Bayesian Equilibrium

A **stationary Bayesian equilibrium** in pure Markov strategies is defined as prices w and r, aggregate profits Π, consumption C, physical capital K and labor supply L, intermediate goods firms' policies $p(z)$, $i(z)$ and value $V(z)$, customer values $M(p)$, U, beliefs $H(z|p)$, an ergodic distribution of customers across firms $G^i(z)$ and mass of unmatched u^i, as well as customer loss and conversion rates $q(p)$ and $e(p)$ such that:

1. Given w, r, Π, consumption C, physical capital K and labor L solve households' problem.
2. Given $w, r, q(p), e(p)$, policies $p(z)$, $i(z)$, and value $V(z)$ solve production firms' problem and $V(\underline{z}) = 0$.
3. Customers' beliefs $H(z|p)$ are consistent with Bayes' law. Customer values $M^i(p)$, U^i, $M(p)$, U satisfy the value equations. Customers switch from p to p' iff $M(p') \geq M(p)$ and from p to unmatched iff $M(p) < U$.
4. Given suppliers' and customers' choices, the masses of matched and unmatched customers $G^i(z)$, u^i, $G(z)$, u evolve consistently, and customer loss and conversion rates $q(p)$ and $e(p)$ are consistent with $G^i(z)$ and $i(z)$.
5. Given w, r, $p(z)$, $G(z)$, u, L, and K, aggregate output Y is implied by the aggregation equation.
6. The labor-market clears: $L = \bar{\ell}\left[u\bar{m}c^{-\sigma}Y + \int_0^\infty \frac{1}{z}p(z)^{-\sigma}YdG(z)\right]$,
   where $\bar{\ell} = \left(\frac{(1-\alpha)(r+\delta_K)}{\alpha w}\right)^\alpha$ is the optimal quantity of labor to produce one unit of output at $z = 1$.
7. The capital market clears: $K = \bar{k}\left[u\bar{m}c^{-\sigma}Y + \int_0^\infty \frac{1}{z}p(z)^{-\sigma}YdG(z)\right]$,
   where $\bar{k} = \left(\frac{\alpha w}{(1-\alpha)(r+\delta_K)}\right)^{1-\alpha}$ is the optimal quantity of capital to produce one unit of output at $z = 1$.
8. The final-goods market clears: $Y = C + (K' - (1-\delta_k)K) + \Gamma$, where $\Gamma = \int_0^\infty s(i(z))dG^i(z)$ is aggregate demand investment.
9. Aggregate profits are given by $\Pi = \int_0^\infty \pi(p(z), z)dG(z)$.

back

Equilibrium Competition for Customers

- Define $G^i(z)$ as distribution of customers and u^i as mass of unmatched (after exit and entry)
  - $D(z) = \int_0^z \frac{i(s)}{I} dG^i(s)$ is the fraction of matches with suppliers of productivity less than z

- Firm **loses** customers whose best contact has productivity higher than z

$$\textbf{quit rate:} \quad q(p) = 1 - e^{-I(1-D(\hat{z}(p)))} \quad \forall p \leq \bar{p}$$

- Firm **attracts** customers that are **matched at lower** z and **do not have better contacts**

$$\textbf{conversion rate:} \quad e(p) = \left(u^i + G^i(\hat{z}(p))\right) e^{-I(1-D(\hat{z}(p)))} \quad \forall p \leq \bar{p}$$

$\Rightarrow$ **price** $p \uparrow \rightarrow$ inferred prod. $\hat{z}(p) \downarrow \rightarrow$ conversion rate $e(p) \downarrow$ & quit rate $q(p) \uparrow \rightarrow$ **growth** $\rho(p, i) \downarrow$

back

Optimal Price and Investment in Demand in Equilibrium

- Condition for optimal price solves differential equation
  - Lowest productivity firm $\underline{z}$ considers only quits to in-house production $\rightarrow M(\bar{p}) = U$ for $\bar{p} = p(\underline{z})$

$$\frac{\partial p(z)}{\partial z} = -\frac{2 + (u^i + G^i(z))i(z)}{1 + (u^i + G^i(z))i(z)} i(z) g^i(z) \frac{W(z)}{\frac{\partial \pi(p,z)}{\partial p}} < 0 \text{ for } z \geq \underline{z}$$

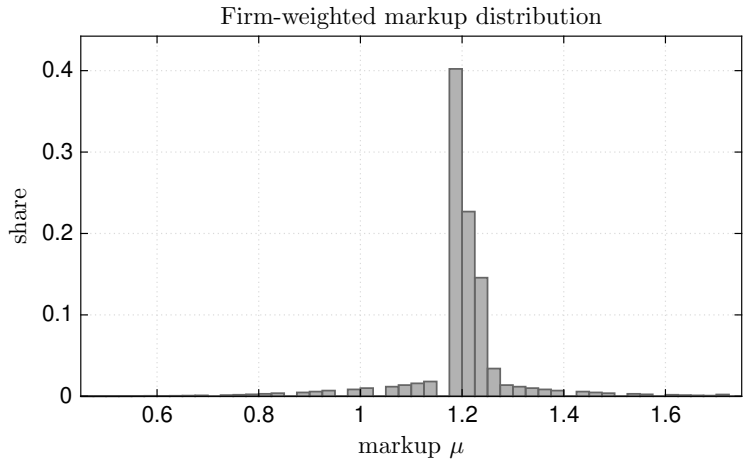
- Optimal investment i solves

$$\frac{\partial s(i(z))}{\partial i} = \left((u^i + G^i(z)) e^{-I(1-D(z))} W(z) \right)$$

$\Rightarrow$ Optimal $p(z)$ and $i(z)$ in equilibrium depend on **distribution of customers**
  - Prices are determined by **local competition** for customers

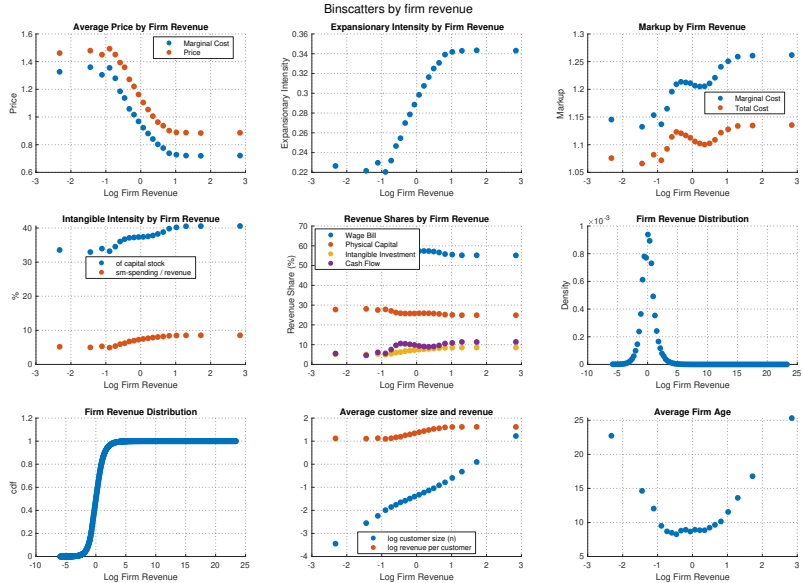
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Markup Distribution (Model)



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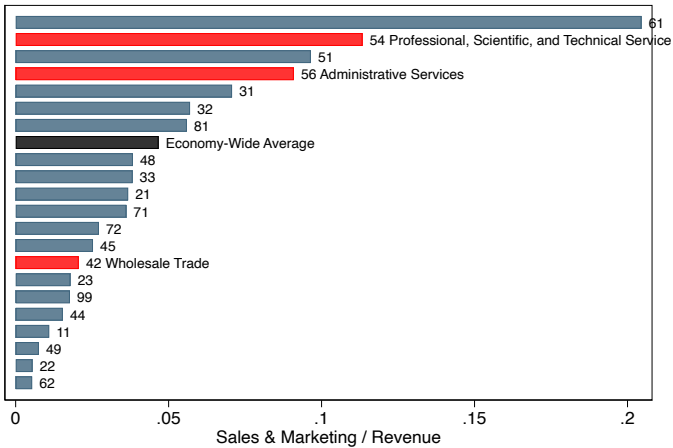
Statistics by Firm Revenue



back

Demand Investment by Industry

- Business-facing industries among sectors with highest sales & marketing intensity

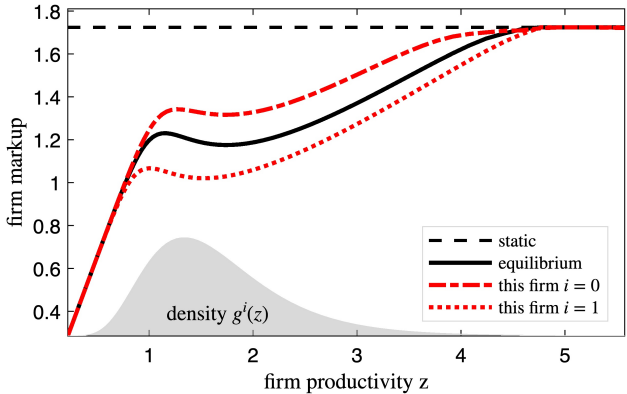


Weighted average by 2-digit NAICS

final-goods firm calibration

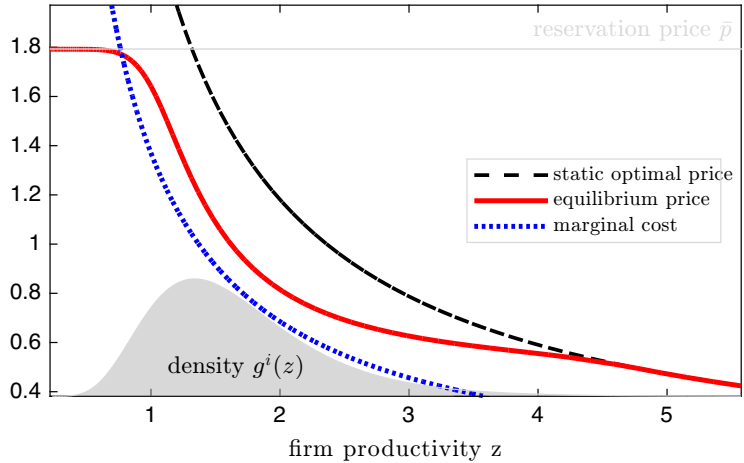
Optimal Markups & Own Demand Investment

- p and i are **complements** to the firm
  - Higher $i \rightarrow$ more contact with new customers $\rightarrow$ higher return to increasing conversion rate $e(p)$



back

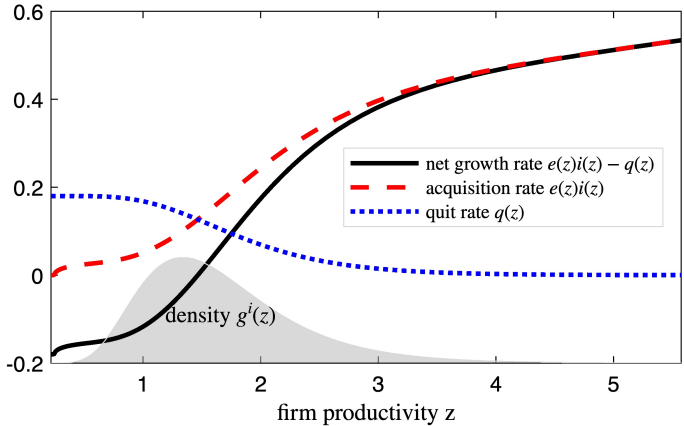
Prices and Marginal Costs



back

Growth in Customer Base

- Low z firms shrink due to quits, high z firms accumulate customers



back

Social Planner: Problem

For given K_0^* and $\{g_{-1}^*(z)\}_{z=0}^{\infty}$, the planner solves

$$\max_{\{C_t^*, L_t^*, K_{t+1}^*, \{i_t^*(z), g_t^{i*}(z), g_t^*(z), \delta_t^*(z)\}_{z=0}^{\infty}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \left\{u(C_t^*) - v(L_t^*)\right\}$$

s.t. $C_t^* + K_{t+1}^* + \int_0^{\infty} s(i_t^*(z)) g_t^{i*}(z) dz \leq Y(K_t^*, L_t^*, \{g_t^*(z)\}) + (1-\delta_k) K_t^* \ \forall t$

$$Y(K_t^*, L_t^*, \{g_t^*(z)\}) = \left(1 + \int_0^{\infty} \left(z^{\sigma-1} - 1\right) g_t^*(z) dz\right)^{\frac{1}{\sigma-1}} (K_t^*)^{\alpha} (L_t^*)^{1-\alpha}$$

$$g_t^*(z) \leq (1 - q_t^*(z) + e_t^*(z) i_t^*(z)) g_t^{i*}(z) \ \forall t, z$$

$$g_t^{i*}(z) \leq \int_0^{\infty} g_{t-1}^*(\hat{z}) \left[(1 - \delta_t^*(z)) f(z|\hat{z}) - \chi(1 - \delta_t^{E*}(z)) f_0(z)\right] d\hat{z} + \chi(1 - \delta_t^{E*}(z)) f_0(z) \ \forall t, z$$

$$\delta_t^*(z) \geq \delta_F, \quad \delta_t^*(z) \leq 1 \ \forall t, z$$

$$\delta_t^{E*}(z) \geq 0, \quad \delta_t^{E*}(z) \leq 1 \ \forall t, z$$

$$e_t^*(z) = \left[1 - \int_z^{\infty} g_t^{i*}(\hat{z}) d\hat{z}\right] e^{-\int_z^{\infty} i_t^*(\hat{z}) g_t^{i*}(\hat{z}) d\hat{z}} \ \forall t, z$$

$$q_t^*(z) = 1 - e^{-\int_z^{\infty} i_t^*(\hat{z}) g_t^{i*}(\hat{z}) d\hat{z}} \ \forall t, z$$

back

Social Planner: Aggregate Market Power

- The planner's optimal choices for labor and capital solve

$$v'(L) = (1-\alpha) Z^M K^\alpha L^{-\alpha} u'(C)$$
$$1 = \beta \left(\alpha Z^M K^{\alpha-1} L^{1-\alpha} + 1 - \delta_K \right)$$

- Substituting the equilibrium wage and interest rate in households' first order condition yields
  - $\mathcal{M}$ is the cost-weighted average markup paid across matches

$$v'(L) = w u'(C) = \frac{1}{\mathcal{M}} (1-\alpha) Z^M K^\alpha L^{-\alpha} u'(C)$$
$$1 = \beta (1+r) = \beta \left(\frac{1}{\mathcal{M}} \alpha Z^M K^{\alpha-1} L^{1-\alpha} + 1 - \delta_K \right)$$

$\Rightarrow$ Aggregate markup $\mathcal{M}$ distorts the optimal choice of labor and capital

back

Social Planner: Intensive Margin Misallocation

- The planner allocates production across matches to maximize aggregate productivity

$$Z^* = \left(u + \int_0^\infty z^{\sigma-1} dG(z) \right)^{\frac{1}{\sigma-1}}$$

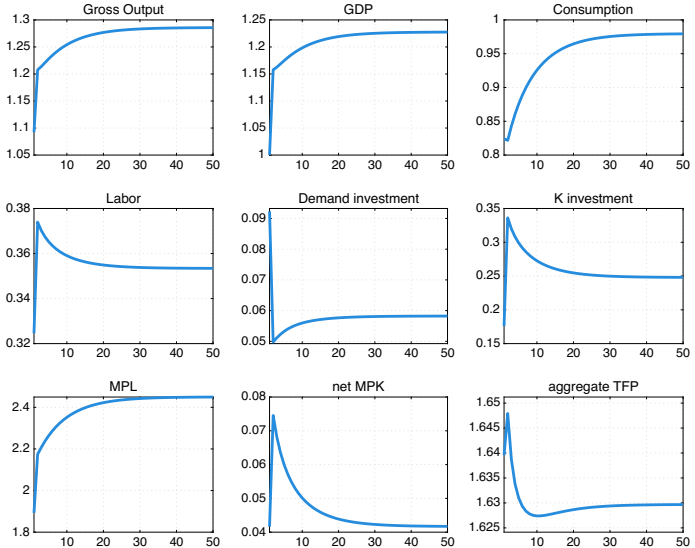
- Aggregate productivity in equilibrium is given as

$$Z^M = \left(u \left(\frac{1}{\mathcal{M}} \right)^{-\sigma} + \int_0^\infty z^{\sigma-1} \left(\frac{\mu(z)}{\mathcal{M}} \right)^{-\sigma} dG(z) \right)^{\frac{1}{\sigma-1}}$$

$\Rightarrow$ Dispersion in markups $\mu(z)$ distorts aggregate productivity

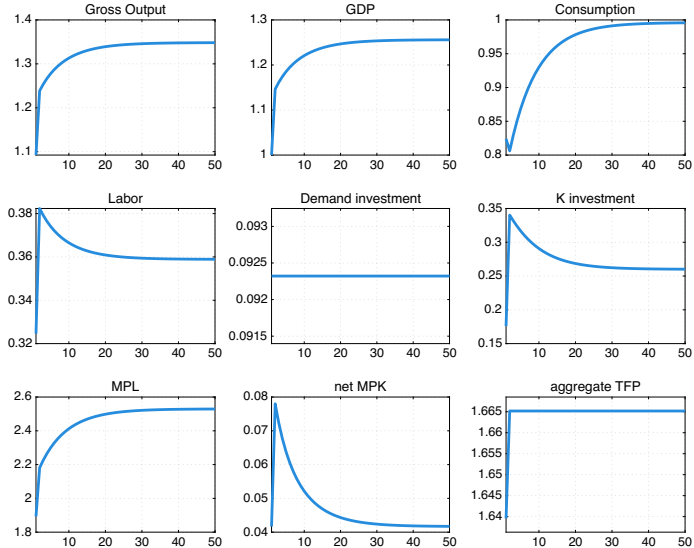
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Transition Planner

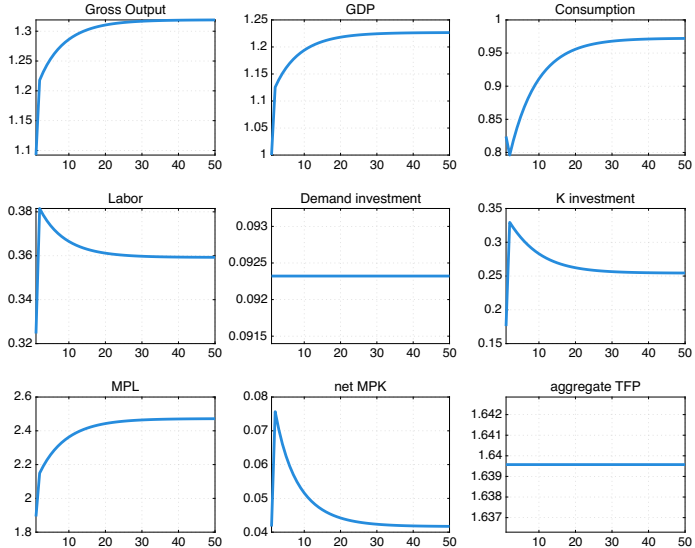


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Transition Planner: Constant $G(z)$



Transition Planner: Constant $G(z)$ & Z



Markups and Misallocation: Role of Customers

- Consider two alternative frameworks:
  1. **Static demand** (Kimball), no extensive margin ($\sim$ Edmond-Midrigan-Xu ('23))
  2. Extensive margin **independent of price**, Kimball at intensive margin ($\sim$ Afrouzi-Drenik-Kim ('25))

  $\Rightarrow$ Calibrated to same level & size-gradient of markups, sales concentration, customer turnover

	Extensive margin	Pricing	Markup level	Markup slope	sd$(\log \mu)$	TFP wedge
Static demand, no customers Edmond-Midrigan-Xu ('23)	no	static	1.197	0.031	0.027	1.84%
Customers unresponsive to price Afrouzi-Drenik-Kim ('25)	yes	static	1.195	0.031	0.053	2.72%
Baseline model Hubmer-Nord ('26)	yes	dynamic	1.197	0.031	0.091	1.56%

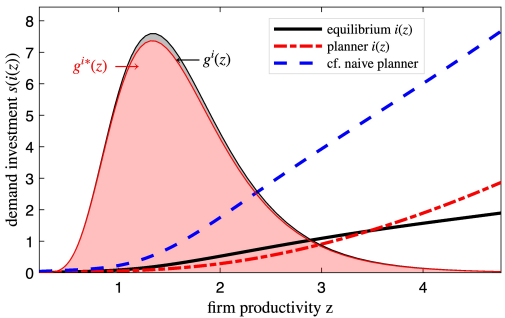
- **Add customers:** het. markups conditional on size $\rightarrow$ dispersion of $\mu(z) \uparrow \rightarrow$ misallocation $\uparrow$
- **Dynamic pricing:** dispersion of $\mu(z) \uparrow\uparrow$, BUT intensive margin elasticity $\downarrow \rightarrow$ misallocation $\downarrow$
  - Higher dispersion due to non-monotone markups and dynamic incentive sustaining $\mu(z) < 1$

$\Rightarrow$ Baseline model yields **low misallocation despite high markup dispersion**

back

Equilibrium vs. Planner Demand Investment: Decomposition

- Both business stealing (over-investment) and differential total surplus (under-investment) large
  - **Counterfactual naive planner** ignores business stealing (& congestion), chooses much higher $i(z)$
  - Net effect $\approx$ 36% overinvestment in equilibrium

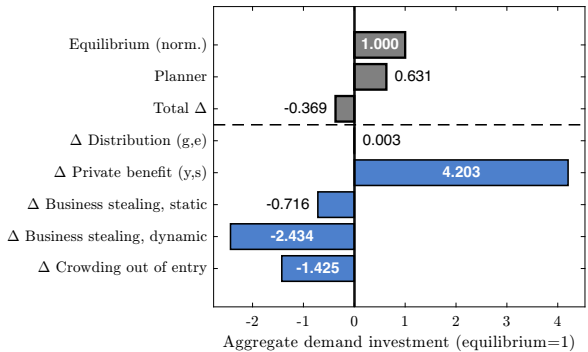


Item	Value
Aggregate spending (equilibrium, norm.)	**100.0%**
Aggregate spending (planner)	**63.1%**
Total Δ (planner $-$ eq.)	-36.9%
Decomposition of total Δ:	
Direct match surplus	$+420.3\%$
Business stealing & congestion	-457.5%
Distribution (g, e)	$+0.3\%$

back

Decomposition: Business Stealing vs. Surplus

- Planner reduces **aggregate** demand investment by **37%** on net
  - **Business stealing** and entry externalities dominate private benefit to customer



back

Demand Investment and the Cost of Markups

- Exercise: counterfactual shift in the level of the contact cost function $\bar{s}$
- Welfare cost of markups increase in $\bar{s}$
  - **Pro-competitive effect**: contact cost $\downarrow \rightarrow$ poaching $\uparrow \rightarrow$ competition in price $\uparrow \rightarrow$ markups $\downarrow$

	Welfare Gain for Planner (% CEV, incl. transition)			
	Total	Aggr. Markup	Markup Dispersion	Dist. of Matches
$\bar{s} - 50\%$	4.66	1.73	1.94	0.99
Baseline calibration	5.14	1.94	2.23	0.97
$\bar{s} + 100\%$	5.72	2.15	2.74	0.82

back

Optimal Product Market Policy

- **Production subsidy**: removes most of the input wedge ⇒ higher K and L
  - Encourages **inefficient demand investm.** → optimal to not fully correct **aggregate markup wedge**

	Production subsidy	Investment tax	Both
Investment tax τ_S	–	33.8%	61.1%
Production subsidy τ_P	13.3%	–	17.1%
Gross output	+19.5%	−5.3%	+15.8%
GDP	+19.3%	−2.7%	+20.2%
Demand investment	+22.4%	−32.8%	−31.6%
Tangible capital	+38.2%	−5.8%	+38.9%
Labor	+9.5%	−1.9%	+9.3%
Consumption	+15.2%	−2.1%	+16.2%
Aggregate TFP	+1.2%	−2.1%	−2.0%
Aggregate Markup	−0.3%	+0.6%	+0.5%
Welfare (CEV) – steady state	+7.0%	−0.7%	+8.0%
Welfare (CEV) – transition	+1.3%	+0.5%	+2.5%

Note: Percent changes relative to baseline no-tax steady state.

back size τ_P τ_S both

Optimal Product Market Policy

- **Investment tax**: corrects **inefficiently high** demand investment
  - Demand investment is **pro-competitive** → optimal to not cut too much

	Production subsidy	Investment tax	Both
Investment tax τ_S	–	33.8%	61.1%
Production subsidy τ_P	13.3%	–	17.1%
Gross output	+19.5%	−5.3%	+15.8%
GDP	+19.3%	−2.7%	+20.2%
Demand investment	+22.4%	−32.8%	−31.6%
Tangible capital	+38.2%	−5.8%	+38.9%
Labor	+9.5%	−1.9%	+9.3%
Consumption	+15.2%	−2.1%	+16.2%
Aggregate TFP	+1.2%	−2.1%	−2.0%
Aggregate Markup	−0.3%	+0.6%	+0.5%
Welfare (CEV) – steady state	+7.0%	−0.7%	+8.0%
Welfare (CEV) – transition	+1.3%	+0.5%	+2.5%

Note: Percent changes relative to baseline no-tax steady state.

back size τ_P τ_S both

Optimal Product Market Policy

- **Both instruments**: higher tax and subsidy, more than additive welfare gain
  - ⇒ Investment tax and production subsidy are **complements**
  - ⇒ Jointly achieve ∼50% of possible welfare gain (85% of agg. markup + extensive margin misalloc.)

	Production subsidy	Investment tax	Both
Investment tax τ_S	–	33.8%	61.1%
Production subsidy τ_P	13.3%	–	17.1%
Gross output	+19.5%	−5.3%	+15.8%
GDP	+19.3%	−2.7%	+20.2%
Demand investment	+22.4%	−32.8%	−31.6%
Tangible capital	+38.2%	−5.8%	+38.9%
Labor	+9.5%	−1.9%	+9.3%
Consumption	+15.2%	−2.1%	+16.2%
Aggregate TFP	+1.2%	−2.1%	−2.0%
Aggregate Markup	−0.3%	+0.6%	+0.5%
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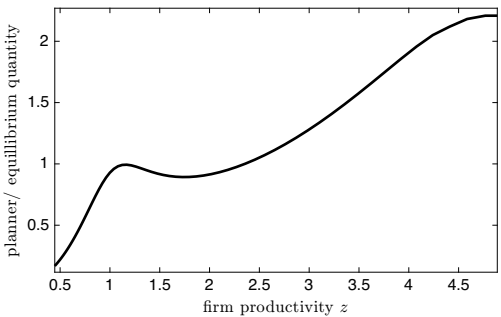
Note: Percent changes relative to baseline no-tax steady state.

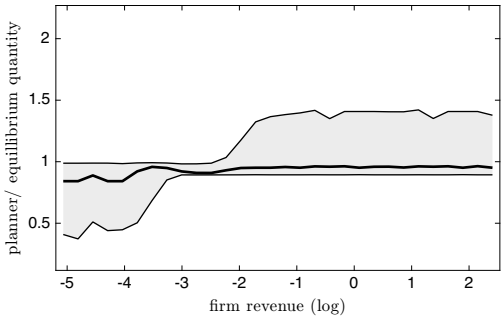
back size τ_P τ_S both

Size Dependent Policies

- Policies to induce full planner solution **difficult to implement**
  - Planner wants to increase sales of firms with high z (high markup)
  - Backward looking # customers $\rightarrow$ total firm revenue poor predictor of z

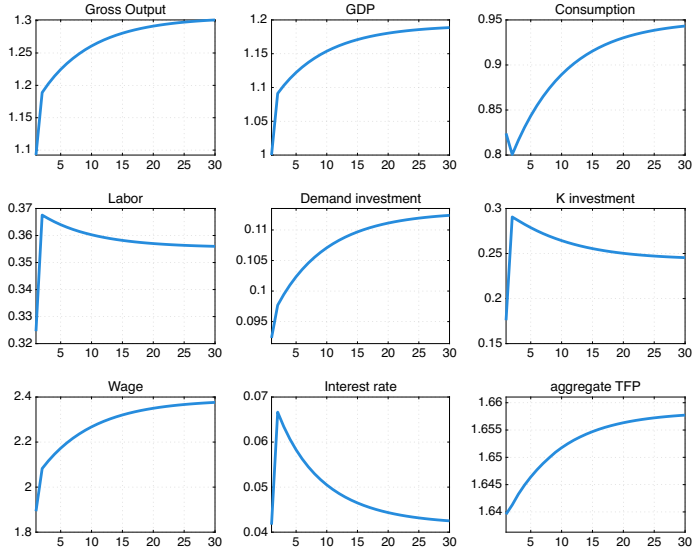
  $\Rightarrow$ **Size dependent policies** ineffective (if condition on total revenue)





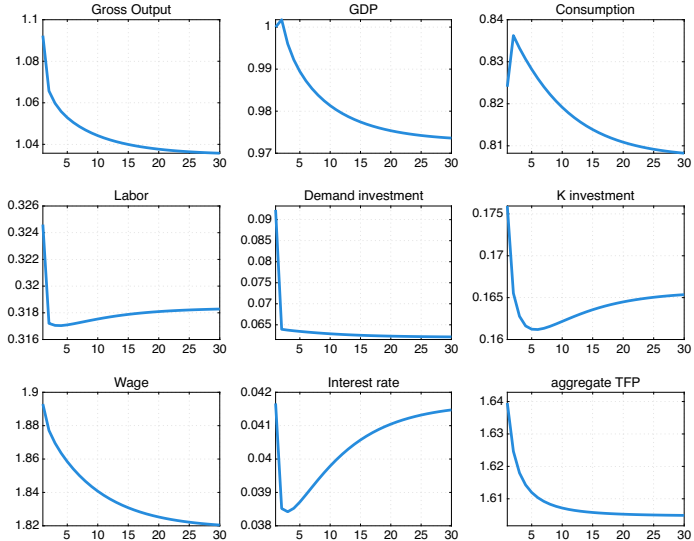
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Transition: Production Subsidy



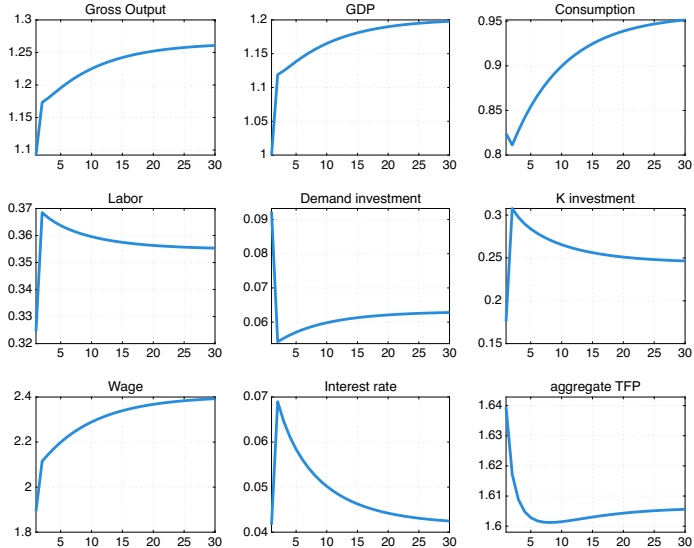
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Transition: Investment Tax



back

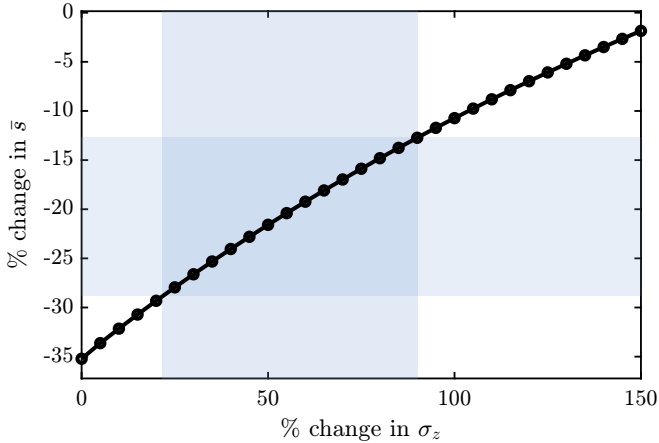
Transition: Joint Policy



back

Rising Aggregate Markup

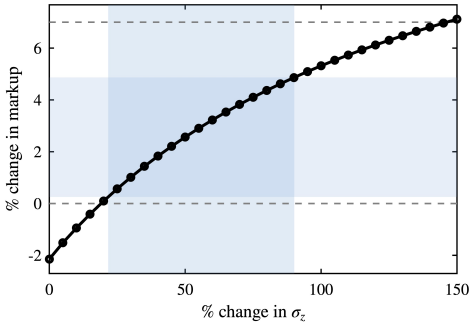
- Explain **observed increase in top 1% sales share** (+25.8%) since 1980s by combination of
  - decrease in contact cost $\bar{s}$
  - rise in firm-level TFP dispersion σ_z



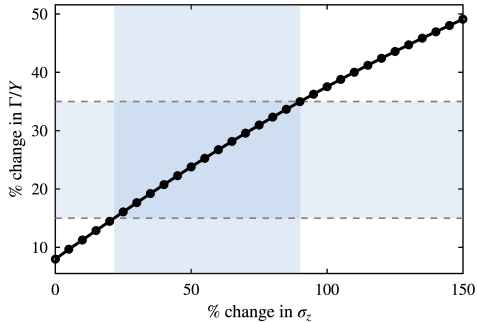
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Rising Aggregate Markup: Results

- The more the observed increase in sales concentration is driven by increased dispersion ($\sigma_z \uparrow$)...
  - ... the larger the increase in aggregate markup (Compustat cost-w.: 0-7pp.)
  - ... while inferred demand investment increase remains plausible



a) Aggregate Markup

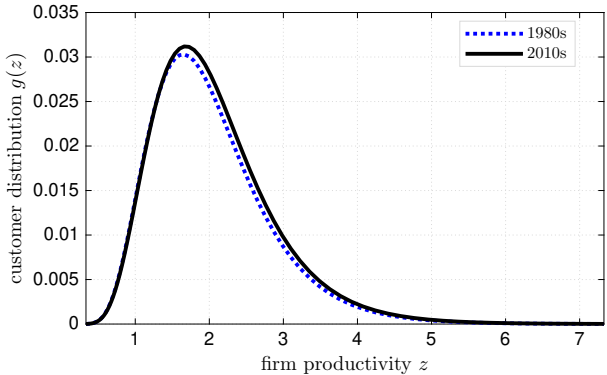


b) Demand Investment

back

Rising Investment in Demand: Distribution of Customers

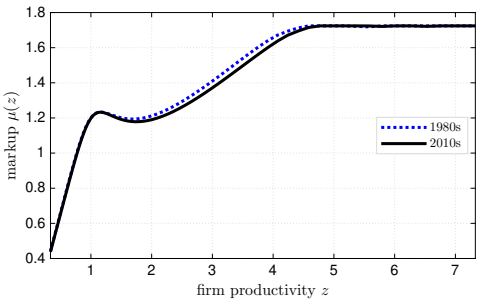
- More matched (fewer unmatched) customers
- Mass reallocated towards more productive firms

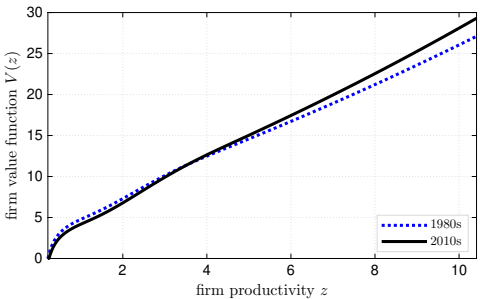


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Rising Investment in Demand: Markups and Firm Values

- Markups decrease in the middle of the distribution (higher $i(z)$ and positive firm mass)
- Firm values decrease at the bottom, increase at the top (change in firm growth)

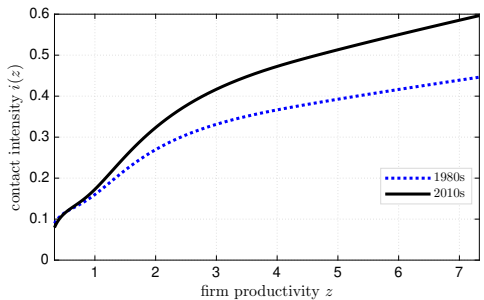


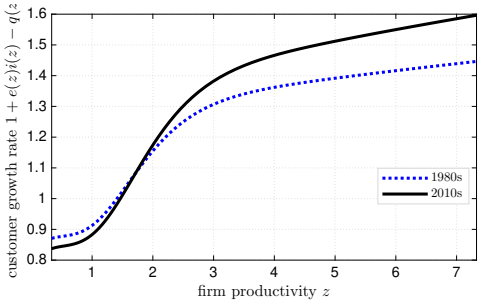


back

Rising Investment in Demand: Investment and Firm Growth

- Investment increases due to lower cost for all but least-productive firms
- Growth increases at the top (higher own $i(z)$) and decreases at the bottom (higher $i(z)$ of others)



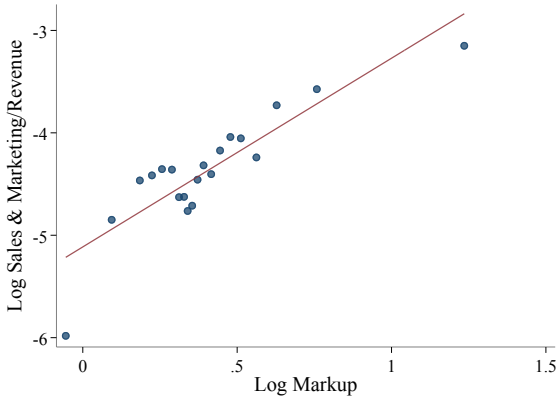


back

Demand Investment vs. Markups in the Data

- **Sales & Marketing** expenses as share of revenue are **increasing in firm markups**
  - Markup $\equiv$ Sales / COGS
- Controlling for industry (NAICS2) and year FEs, weighted by COGS
- Elasticity: **1.84**
  - Targeted in calibration

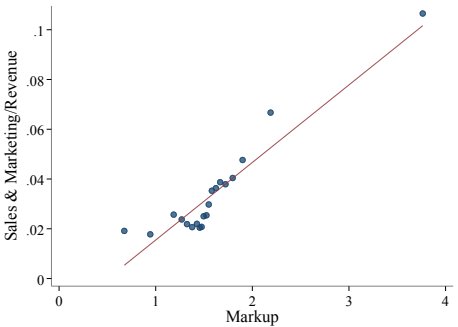
Data: He, Mostrom, Sufi ('25) / Capital IQ



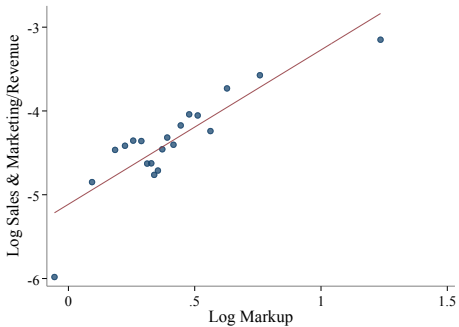
back table level spec

Demand Investment vs. Markups

- Capital IQ: Sales & Marketing expenses as share of firm revenue are **increasing** in firm markups
  - Controlling for industry (NAICS-2) and year fixed effects



(a) Level Specification



(b) Log-Log Specification

back

Demand Investment vs. Markups

	(1)	(2)	(3)	(4)	(5)	(6)
Log markup ($\ln \frac{Revenue}{COGS}$)	1.295***	2.195***	0.880***	**1.843*** **	1.011***	1.720***
	(0.256)	(0.260)	(0.193)	(0.379)	(0.143)	(0.256)
Observations	30,674	30,674	30,674	30,674	30,674	30,674
R^2	0.095	0.176	0.162	0.309	0.300	0.339
Fixed Effects	—	—	Year+Ind	Year+Ind	Year+Ind	Year+Ind
Weights	—	Cost	—	Cost	—	Cost
Controls	—	—	—	—	Age+Size	Age+Size

Data source: He, Mostrom, Sufi (2025); Capital IQ. Dependent variable: log ratio of sales and marketing spending to revenue. Robust standard errors clustered at industry level. *** p<0.01, ** p<0.05, * p<0.1.

back

Demand Investment vs. Markups

	(1)	(2)	(3)	(4)	(5)	(6)
Markup ($\frac{Revenue}{COGS}$)	0.029**	0.037***	0.004	0.031***	0.017*	0.029***
	(0.014)	(0.008)	(0.015)	(0.008)	(0.010)	(0.008)
Observations	30,674	30,674	30,674	30,674	30,674	30,674
R^2	0.005	0.135	0.042	0.166	0.266	0.214
Fixed Effects	—	—	Year+Ind	Year+Ind	Year+Ind	Year+Ind
Weights	—	Cost	—	Cost	—	Cost
Controls	—	—	—	—	Age+Size	Age+Size

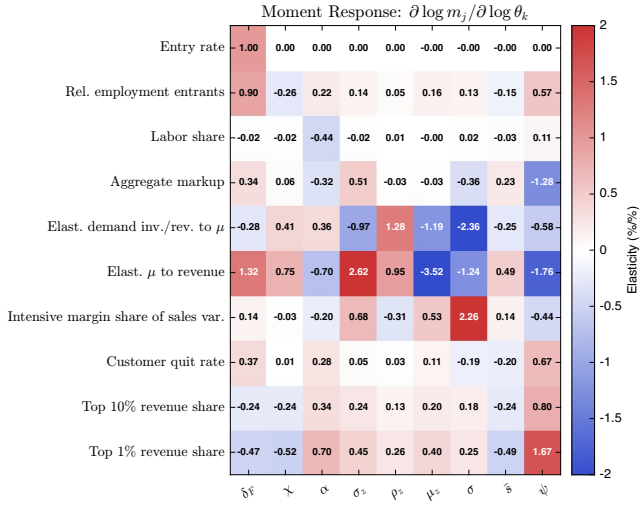
Data source: He, Mostrom, Sufi (2025); Capital IQ. Dependent variable: ratio of sales and marketing spending to revenue. Robust standard errors clustered at industry level. *** p<0.01, ** p<0.05, * p<0.1.

back

Parameter-Moment Sensitivity: Moment-Response Elasticity

back

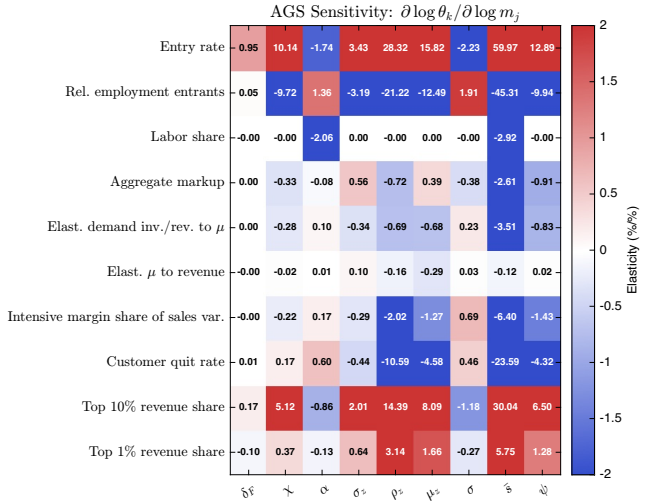
- $E_{kj} = \partial \log m_j / \partial \log \theta_k$: how each moment responds to parameter changes



Parameter-Moment Sensitivity: Parameter Sensitivity Λ

back

- $\Lambda = (EE^\top)^{-1}E$: how estimated $\hat{\theta}_k$ responds to change in data moment m_j Andrews, Gentzkow, Shapiro ('17)



Robustness: Calibration Parameters

back to calibration

	Baseline	Quitrate	Int. Margin		Agg. Markup	
Modified target:		↑ 20%	↓ 15%	↑ 35%	↓ 15%	↑ 35%
Parameters (9)						
Exogenous firm exit rate (δ_F)	0.081	0.085	0.080	0.082	0.080	0.082
Entrants per unmatched customer (χ)	0.078	0.236	0.058	0.113	0.080	0.099
Capital elasticity (α)	0.327	0.326	0.324	0.332	0.344	0.295
Log TFP: unconditional dispersion (σ_z)	0.387	0.464	0.383	0.398	0.279	0.519
Log TFP: auto-correlation (ρ_z)	0.361	0.350	0.222	0.415	0.354	0.509
Log TFP: mean (μ_z)	0.409	0.603	0.360	0.448	0.264	0.688
Intensive-margin elasticity (σ)	2.381	2.380	1.939	2.814	3.042	2.003
Contact cost: level ($\bar{s}$)	39.792	25.056	59.757	30.032	89.442	74.259
Contact cost: curvature (ψ)	2.895	3.074	3.223	2.671	4.000	2.936
Targeted moments (10)						
Firm entry/exit rate	0.081	0.086	0.080	0.082	0.080	0.082
Relative employment of entrants	0.526	0.552	0.521	0.533	0.524	0.534
Aggregate labor share	0.614	0.615	0.616	0.611	0.615	0.618
Aggregate markup	0.250	0.249	0.251	0.250	0.150	0.351
Elasticity of $\frac{\text{demand investment}}{\text{revenue}}$ to markup	1.849	1.838	1.848	1.851	1.848	1.852
Elasticity of markup to revenue	0.031	0.032	0.031	0.031	0.031	0.031
Sales variation: intensive-margin share	0.262	0.266	0.150	0.355	0.263	0.263
Customer quit rate (% sales)	0.152	0.189	0.152	0.153	0.153	0.154
Top 10% sales share	0.676	0.702	0.667	0.691	0.674	0.682
Top 1% sales share	0.431	0.441	0.431	0.430	0.430	0.432

Robustness: Social Planner & Optimal Taxation

back to policies

	Baseline	Quitrate	Int. Margin Sh.		Agg. Markup	
Modified target:		↓ 20%	↓ 15%	↑ 35%	↓ 15%	↑ 35%
Social Planner: % Δ relative to equilibrium (steady state)						
Gross Output	17.72	20.70	15.83	19.87	11.44	28.10
GDP	22.77	25.84	21.03	24.73	14.24	33.16
Demand investment	-36.92	-27.49	-47.46	-28.14	-46.46	-12.51
Tangible Capital	40.91	46.34	37.52	44.45	24.71	64.41
Labor	8.86	9.76	8.16	9.51	5.62	13.54
Consumption	18.90	21.47	17.55	20.44	11.79	27.54
Aggregate TFP	-0.60	0.12	-0.92	-0.14	-0.35	1.15
Social Planner: % Δ welfare relative to equilibrium (incl. transition)						
Total	5.14	5.75	5.13	5.29	3.10	6.92
Decomposition:						
Aggregate markup	1.94	2.23	1.75	2.12	0.79	3.44
Markup dispersion	2.23	2.37	2.16	2.16	1.52	2.91
Demand investment	0.97	1.15	1.22	1.02	0.79	0.58
Optimal Taxation						
Demand investment tax only						
Optimal demand investment tax	33.82	28.96	47.27	26.55	59.57	7.82
Welfare gain (% Δ)	0.46	0.36	0.69	0.34	0.51	0.05
Production subsidy only						
Optimal production subsidy	13.26	14.50	12.38	14.00	8.66	19.28
Welfare gain (% Δ)	1.28	1.55	1.07	1.47	0.52	2.76
Demand investment tax + production subsidy						
Optimal demand investment tax	61.10	58.10	75.30	53.65	84.58	34.66
Optimal production subsidy	17.05	18.31	16.30	17.66	11.25	22.52
Welfare gain (% Δ)	2.45	2.68	2.47	2.52	1.36	3.48

Robustness: Counterfactual Analysis (1980s vs. 2010s)

back to comp. statics

	Baseline	Quitrate	Int. Margin Sh.		Agg. Markup	
Modified target:		↓ 20%	↓ 15%	↑ 35%	↓ 15%	↑ 35%
Targeted moments, % △ across steady states (1980s vs. 2010s)						
Agg. demand investment / Gross output	15.00	15.00	15.00	15.00	15.00	15.00
Top 1% sales share	25.80	25.80	25.80	25.80	25.80	25.80
Calibrated % in contact technology, for fixed $i(z)$						
Average demand investment	-31.03	-27.99	-40.33	-24.96	-45.83	-25.89
P75/25 demand investment	-8.28	-12.57	-5.77	-10.13	-8.27	-9.09
Untargeted moments, % △ across steady states (1980s vs. 2010s)						
GDP	4.57	4.85	4.91	4.33	3.64	6.02
Aggregate TFP	3.11	3.52	3.15	3.12	2.08	4.44
Avg. customer quit rate	8.77	7.53	10.51	7.44	9.08	7.47
Aggregate markup	-0.36	-0.14	-0.58	-0.14	-0.59	-0.34
Untargeted moments, % △ simulated firm panel across transition path						
Intangible capital intensity (true)	-12.45	-10.33	-14.19	-10.95	-12.89	-10.32
Intangible capital intensity (cap.)	8.47	12.84	5.98	9.93	9.19	10.00

Validating Customer Surplus and Business Stealing

back

- Are our estimates of customer surplus and business stealing externalities reasonable?

- **Customer surplus:** Model matches estimated benefit of additional supplier → **static surplus** ✓
  - Estimated from cost reduction following addition/loss of supplier → maps into static benefit $\eta(p)$
  - Static customer surplus as share of expenditure: model 30% vs. data 28% (Baqaee-Burstein-Duprez-Farhi ('25))

- **Business stealing (proxy):** Does demand investment increase **total matches**?
  - Fraction of a firm's new matches that displace existing matches in model: 71%
  - Empirical estimates from advertising: 72–100% (✓) (Garthwaite ('14), Hartmann-Klapper ('18), Sinkinson-Starc ('19))
  - Model: Displacement partially offset by gain in productivity across matches

- Dynamic surplus / business stealing implicitly disciplined by **persistence of matches** ✓

Rise in Demand Investment

- Since 1980s, **share of sales and marketing spending in revenue** has increased
  - Kaplan-Zoch ('24): Share & average wage of expansionary workers has increased (1980-2015)
  - Bronnenberg-Dube-Syverson ('22): Marketing managers share of payroll increased ∼25% (2005-19)
  - Greenwood-Ma-Yorukoglu ('24): $\frac{\text{advertising}}{\text{GDP}}$ has increased by $\sim 35\%$ (1980-2019)
  - KLEMS: nominal $\frac{\text{brand value investment}}{\text{GDP}}$ has increased by $\sim 15\%$ (1985-2020)

- Candidate driver: **Technological change**
  - Calibrate change in **level and elasticity** of contact cost function $s(\cdot)$
  - **Targets**: KLEMS increase in agg. spending, increase in top 1% sales share from Autor et al. ('20)

back

Rise in Demand Investment: 1980s vs. 2010s

- Model matches targets with a **decline in level and curvature of contact cost function**

		1980s	2010s	% Change
$Y_{Dem.Inv.}/Y_{gross}$	(target $+15\%$)	0.073	0.085	$+15.0\%$
Top 1% sales share	(target $+25.8\%$)	0.342	0.431	$+25.8\%$
Avg. demand investment	(fix $i(z)$)	0.326	0.225	-31.0%
P75/P25 demand investment	(fix $i(z)$)	3.430	3.146	-8.3%
GDP		1.000	1.046	$+4.6\%$
Aggregate TFP		1.590	1.640	$+3.1\%$
Aggregate markup		1.201	1.197	-0.4%
Intangible capital intensity		0.403	0.369	-8.6%
Customer quit rate		0.140	0.152	$+8.8\%$

Notes: Quantities normalized so that $GDP = 1$ in the 1980s steady state.

▸ distribution of customers ▸ markups and firm values ▸ $i(z)$ and growth ▸ robustness ▸ markup increase ◂ back

Rise in Demand Investment: 1980s vs. 2010s

- Increased demand investment shifts matches to higher z, **improving TFP & GDP**

		1980s	2010s	% Change
$Y_{Dem.Inv.}/Y_{gross}$	(target $+15\%$)	0.073	0.085	$+15.0\%$
Top 1% sales share	(target $+25.8\%$)	0.342	0.431	$+25.8\%$
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▸ distribution of customers ▸ markups and firm values ▸ $i(z)$ and growth ▸ robustness ▸ markup increase ◂ back

Rise in Demand Investment: 1980s vs. 2010s

- Little change in aggregate markup $\bar{\mu} \rightarrow$ **rise in concentration, no decline in competition**
  - Offsetting effects: at **firm level** $\mu(z) \downarrow$ (competition ↑), but **reallocation to high-markup firms** $\bar{\mu} \uparrow$

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▸ distribution of customers ▸ markups and firm values ▸ $i(z)$ and growth ▸ robustness ▸ markup increase ◂ back

Rise in Demand Investment: 1980s vs. 2010s

- **Intangible capital intensity declines** despite higher investment

		1980s	2010s	% Change
$Y_{Dem.Inv.}/Y_{gross}$	(target $+15\%$)	0.073	0.085	$+15.0\%$
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Intangible capital intensity		0.403	0.369	-8.6%
Customer quit rate		0.140	0.152	$+8.8\%$

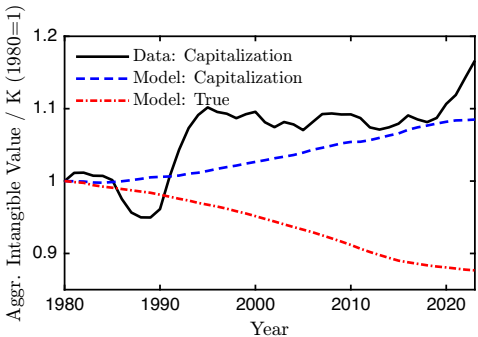
Notes: Quantities normalized so that $GDP = 1$ in the 1980s steady state.

▸ distribution of customers ▸ markups and firm values ▸ $i(z)$ and growth ▸ robustness ▸ markup increase ◂ back

Intangible Capital: Capitalization vs. True Value

- As in the data, model features **increased spending on demand investment**
  - Simulation based on linear change in $s(\cdot)$ parameters from 1980s to 2010s steady state

- **Capitalized** intangible value ↑ vs. **true intangible intensity** ↓
  - Capitalization under constant (average) depreciation rate
  - Difference reflects **equilibrium effect** of increased poaching → **endogenous "depreciation"**



Intangible Capital: Heterogeneous Firm Valuations

- Changes in contact technology do not affect firms equally
  - Increased poaching hurts less productive firms, but allows more productive firms to grow faster

$\Rightarrow$ Firm intangible value **fans out**: increases at top of distribution relative to bottom
  - Qualitatively consistent with evidence (e.g., Eeckhout '25)

